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COMMON QUARTERLY EXAMINATION – SEPTEMBER 2026

12 - STD MATHEMATICS

Time : 3.00 Hrs **Maximum Marks: 90**

Instructions: 1. Check the question paper for fairness of printing. If there is any Lack of fairness,inform the Hall Supervisor Immediately
2.Use Blue or Black Ink to write and underline and pencil to draw Diagrams.

PART-I

Note: I) Answer all the questions. 20x1=20
II) Choose the most appropriate answer from the given four alternatives And write the option code and the corresponding answer.

1. If $\rho(A)=\rho([A\,|\,B])$, then the system $AX=B$ of linear equations is
(1) consistent and has a unique solution (2) consistent
(3) consistent and has infinitely many solution (4) inconsistent

2. If $A=\begin{bmatrix}\frac{3}{5} & \frac{4}{5}\\ x & \frac{3}{5}\end{bmatrix}$ and $A^T=A^{-1}$, then the value of x is
(1) $\frac{-4}{5}$ (2) $\frac{-3}{5}$ (3) $\frac{3}{5}$ (4) $\frac{4}{5}$

3. If A is an orthogonal matrix then
(1) $|A|=\pm 2$ (2) $|A|=0$ (3) $|A|=\pm 1$ (4) $|A|=\pm 3$

4. If $|A|\neq 0$, then which of the following is not true?
(1) $(A^2)^{-1}=(A^{-1})^2$ (2) $(A^T)^{-1}=(A^{-1})^T$
(3) $A^{-1}=|A|^{-1}$ (4) $(\text{adj}\,A)^T=(\text{adj}\,A^T)$

5. If $z=x+iy$ is a complex number such that $|z+2|=|z-2|$, then the locus of z is
(1) real axis (2) imaginary axis (3) ellipse (4) circle

6. The solution of the equation $|z|-z=1+2i$ is
(1) $\frac{3}{2}-2i$ (2) $-\frac{3}{2}+2i$ (3) $2-\frac{3}{2}i$ (4) $2+\frac{3}{2}i$

7. If z_1, z_2 are two complex numbers such that $|z_1|=|z_2|=1$, then the minimum value of $|z_1+1|+|z_2+1|+|z_1z_2+1|$ is (1) –1 (2) –3 (3) 2 (4) –2

8. If $z=\frac{3+4i}{2-3i}$, the complex number ω which satisfies the equation $z\omega=1$ is
(1) $\omega=\frac{6+17i}{25}$ (2) $\omega=\frac{-6-17i}{25}$ (3) $\omega=\frac{-6-17i}{5}$ (4) $\omega=\frac{6+17i}{5}$

9. The polynomial x^3-kx^2+9x has three real zeros if and only if, k satisfies
(1) $|k|\leq 6$ (2) $k=0$ (3) $|k|>6$ (4) $|k|\geq 6$

10. A polynomial $p(x)$ of degree n is said to be a reciprocal polynomial if
(1) either $p(x)=x^n p\left(\frac{1}{x}\right)$ or $p(x)=-x^n p\left(\frac{1}{x}\right)$ (2) $p(x)=x^n p\left(\frac{1}{x}\right)$ and $p(x)=-x^n p\left(\frac{1}{x}\right)$
(3) either $p(x)=p\left(\frac{1}{x}\right)$ or $p(x)=p\left(-\frac{1}{x}\right)$ (4) $p(x)=p\left(\frac{1}{x}\right)$ and $p(x)=p\left(-\frac{1}{x}\right)$

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11. If $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3}$; then $\cos^{-1}x+\cos^{-1}y$ is equal to

(1) $\frac{2\pi}{3}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{6}$ (4) π

12. If $\cot^{-1}2$ and $\cot^{-1}3$ are two angles of a triangle, then the third angle is

(1) $\frac{\pi}{4}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{3}$

13. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(1) $2ab$ (2) ab (3) $\sqrt{ab}$ (4) $\frac{a}{b}$

14. The radius of the circle $3x^2+by^2+4bx-6by+b^2=0$ is

(1) 1 (2) 3 (3) $\sqrt{10}$ (4) $\sqrt{11}$

15. If $P(x,y)$ be any point on $16x^2+25y^2=400$ with foci $F_1(3,0)$ and $F_2(-3,0)$ then PF_1+PF_2 is

(1) 8 (2) 6 (3) 10 (4) 12

16. The line PP' is a focal chord of the parabola $y^2=8x$ and if the coordinates of P are $(18,12)$ then the coordinates of P' is

(1) $\left(\frac{2}{9},\frac{-4}{3}\right)$ (2) $\left(\frac{-2}{9},\frac{-4}{3}\right)$ (3) $\left(\frac{-2}{9},\frac{-4}{3}\right)$ (4) $\left(\frac{2}{3},\frac{-4}{9}\right)$

17. If $\vec{a},\vec{b},\vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a},\vec{b},\vec{c}]=3$, then $\{[\vec{a}\times\vec{b},\ \vec{b}\times\vec{c},\ \vec{c}\times\vec{a}]\}^2$ is equal to

(1) 81 (2) 9 (3) 27 (4) 18

18. If the distance of the point $(1,1,1)$ from the origin is half of its distance from the plane $x+y+z+k=0$, then the values of k are

(1) ±3 (2) ±6 (3) $-3,9$ (4) $3,-9$

19. If $|\vec{a}|=2$, $|\vec{b}|=5$, and $|\vec{a}\times\vec{b}|=8$, then the value of $(\vec{a}\cdot\vec{b})^2$ is:

(1) 16 (2) 25 (3) 36 (4) 64

20. If $\vec{a},\vec{b},\vec{c}$ are three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, then the magnitude $|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|$ is equal to:

(1) $\sqrt{3}$ (2) $\frac{3\sqrt{3}}{2}$ (3) 3 (4) $\frac{\sqrt{3}}{2}$

PART-II

Answer any seven questions. Question no. 30 is compulsory 7x2=14

21. Find the rank of the matrix $\begin{bmatrix}1 & -2 & -1 & 0\\ 3 & -6 & -3 & 1\end{bmatrix}$

22. If A and B are any two non-singular square matrices of order n, then $\text{adj}(AB)=(\text{adj}\ B)(\text{adj}\ A)$.

23. Obtain the Cartesian form of the locus of $z=x+iy$ in $\bar{z}=z^{-1}$.

24. Show that, if p,q,r are rational, the roots of the equation $x^2-2px+p^2-q^2+2qr-r^2=0$ are rational.

25. Solve $\sin^{-1}x>\cos^{-1}x$

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26. Obtain the equation of the circles with radius 5 cm and touching x-axis at the origin in general form.

27. The equation $y=\frac{1}{32}x^2$ models cross sections of parabolic mirrors that are used for solar energy. There is a heating tube located at the focus of each parabola; how high is this tube located above the vertex of the parabola?

28. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors represented by concurrent edges of a parallelepiped of volume 4 cubic units, find the value of $(\vec{a}+\vec{b})\cdot(\vec{b}\times\vec{c})+(\vec{b}+\vec{c})\cdot(\vec{c}\times\vec{a})+(\vec{c}+\vec{a})\cdot(\vec{a}\times\vec{b})$.

29. Show that the points $(2,3,4),(-1,4,5)$ and $(8,1,2)$ are collinear.

30. If $i=\sqrt{-1}$ Find the value of $\left(\frac{1+i}{\sqrt{2}}\right)^{2024}+\left(\frac{1-i}{\sqrt{2}}\right)^{2024}$

PART-III

Answer any seven questions. Question no. 40 is compulsory 7x3=21

31. If $A=\begin{bmatrix}3 & 2\\ 7 & 5\end{bmatrix}$ and $B=\begin{bmatrix}-1 & -3\\ 5 & 2\end{bmatrix}$, verify that $(AB)^{-1}=B^{-1}A^{-1}$.

32. A fish tank can be filled in 10 minutes using both pumps A and B simultaneously. However, pump B can pump water in or out at the same rate. If pump B is inadvertently run in reverse, then the tank will be filled in 30 minutes. How long would it take each pump to fill the tank by itself? (Use Cramer's rule to solve the problem).

33. Find the quotient $\frac{2\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)}{4\left(\cos\left(\frac{-3\pi}{2}\right)+i\sin\left(\frac{-3\pi}{2}\right)\right)}$ in rectangular form.

34. The complex numbers u,v, and w are related by $\frac{1}{u}=\frac{1}{v}+\frac{1}{w}$. If $v=3-4i$ and $w=4+3i$, find u in rectangular form.

35. If p is real, discuss the nature of the roots of the equation $4x^2+4px+p+2=0$, in terms of p.

36. Find the value of $\cos\left[\frac{1}{2}\cos^{-1}\left(\frac{1}{8}\right)\right]$

37. The maximum and minimum distances of the Earth from the Sun respectively are 152×10^6 km and 94.5×10^6 km. The Sun is at one focus of the elliptical orbit. Find the distance from the Sun to the other focus.

38. Find the torque of the resultant of the three forces represented by $-3\hat{i}+6\hat{j}-3\hat{k}$, $4\hat{i}-10\hat{j}+12\hat{k}$ and $4\hat{i}+7\hat{j}$ acting at the point with position vector $8\hat{i}-6\hat{j}-4\hat{k}$, about the point with position vector $18\hat{i}+3\hat{j}-9\hat{k}$.

39. Find the coordinates of the point where the straight line $\vec{r}=\left(2\hat{i}-\hat{j}+2\hat{k}\right)+t\left(3\hat{i}+4\hat{j}+2\hat{k}\right)$ intersects the plane $x-y+z-5=0$.

40. Find the equation of the hyperbola whose centre is $(1,-1)$, transverse axis is parallel to the y-axis of length 6, and eccentricity $e=2$.

PART-1V

Answer all the questions: 7x5=35

41a) Solve the following system of equations, using matrix inversion method:
$2x_1+3x_2+3x_3=5,\quad x_1-2x_2+x_3=-4,\quad 3x_1-x_2-2x_3=3.$

b) Solve the equation $z^3+8i=0$, where $z\in\mathbb{C}$.

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42 a) Test for consistency of the following system of linear equations and if possible solve: $x+2y-z=3,\ 3x-y+2z=1,\ x-2y+3z=3,\ x-y+z+1=0$.

(OR)

b) Prove by vector method that $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$.

43 a) If $2+i$ and $3-\sqrt{2}$ are roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$, find all roots.

(OR)

b) A rod of length $1.2m$ moves with its ends always touching the coordinate axes. The locus of a point P on the rod, which is $0.3m$ from the end in contact with x-axis is an ellipse. Find the eccentricity.

44 a) Find the centre, foci, and eccentricity of the hyperbola $11x^2-25y^2-44x+50y-256=0$

(OR)

b) Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point $(1,-2,4)$ and perpendicular to the plane $x+2y-3z=11$ and parallel to the line $\frac{x+7}{3}=\frac{y+3}{-1}=\frac{z}{1}$.

45 a) If $2\cos\alpha=x+\frac{1}{x}$ and $2\cos\beta=y+\frac{1}{y}$, show that

(i) $\frac{x^m}{y^n}-\frac{y^n}{x^m}=2i\sin(m\alpha-n\beta)$ (ii) $x^m y^n+\frac{1}{x^m y^n}=2\cos(m\alpha+n\beta)$.

(OR)

b) Prove by vector method that the perpendiculars (attitudes) from the vertices to the opposite sides of a triangle are concurrent.

46a) Find the equation of the ellipse whose eccentricity is $\frac{1}{2}$, one of the foci is $(2,3)$ and a directrix is $x=7$. Also find the length of the major and minor axes of the ellipse.

(OR)

b) If $a_1, a_2, a_3, \ldots a_n$ is an arithmetic progression with common difference d,

prove that $\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\ldots+\tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right)\right]=\frac{a_n-a_1}{1+a_1a_n}$.

47 a) A semielliptical archway over a one-way road has a height of $3m$ and a width of $12m$. The truck has a width of $3m$ and a height of $2.7m$. Will the truck clear the opening of the archway?

(OR)

b) Find the non-parametric form of vector equation, and Cartesian equations of the plane passing through the points $(2,2,1)$, $(9,3,6)$ and perpendicular to the plane $2x+6y+6z=9$.